

HW2: Robust Regression

Group : G4

Contributors

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Your tasks are:

Download the Google sheet and answer all the questions. Please download as a PDF file and rename it from 'Lab2.pdf' to 'Lab2_G1.pdf'.

Go to Github [Lab2](#) and [Google Doc](#)

1. We have partially implemented a framework to find outliers. Your task is to complete the following functions:
 - `'H_matrix_leverage'` that compute the H-matrix and find the diagonal elements in the matrix
 - `'CookDistance'` to compute the Cook's distance for all $j = 1 \dots m$ elements.

The two outlier analyses are provided in [outlier_diagnostics.ipynb](#). This notebook provides:

- Tools and packages
 - Download data from [kaggle](#)
 - Data processing
 - For more information, see [Lab2/readme.md](#)
2. Point out the differences between H-matrix and Cook's distance in the algorithm from a theoretical point of view. Mainly, you should be able to verify...
 - a. Are the outliers identified by the two methods intersect?
 - b. Are there any outliers identified that have been chosen differently by each of the methods?
 - c. How is the performance improved by removing the outliers identified by H-matrix versus the Cook's distance?

Please answer these questions using your results to provide the discussion. Make sure that every point that you made is supported by the evidence and the data.

Question 1

We have partially implemented a framework to find outliers. Your task is to complete the following functions:

- `H_matrix_leverage` that compute the H-matrix and find the diagonal elements in the matrix

```
def H_matrix_leverage(X_train):
    """
    Compute leverage scores (diagonal of the hat matrix H) for training inputs.

    Parameters
    -----
    X_train : numpy array of shape (n_samples, n_features)
        Feature matrix (ONLY X, not including target y)

    Returns
    -----
    H_diag : numpy array of shape (n_samples,)
        Leverage score for each sample (diagonal elements of hat matrix)
    """

    n_samples, n_feats = X_train.shape

    X = np.hstack([np.ones((n_samples, 1)), X_train])
    XtX_inv = np.linalg.pinv(X.T @ X)
    XA = X @ XtX_inv
    H_diag = np.sum(XA * X, axis=1)

    return H_diag
```

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- `CookDistance` to compute the Cook's distance for all $j = 1 \dots m$ elements.

```
def CookDistance(e_train, mse_train, H_diag, num_feats):  
    """  
    Compute Cook's Distance for each training sample.  
  
    Parameters  
    -----  
    e_train    : array of residuals (n_samples,)  
    mse_train  : mean squared error of the regression model  
    H_diag     : leverage values (diagonal of hat matrix)  
    num_feats  : number of input features  
  
    Returns  
    -----  
    Cook_D : array of Cook's Distance values (n_samples,)  
    """  
    n = num_feats + 1 # To remind: '+1' because of intercept term  
  
    # Small value to avoid division by zero  
    eps = 1e-12  
  
    numerator = e_train**2 * H_diag  
    denominator = n * mse_train * (1 - H_diag)**2  
    Cook_D = numerator / denominator  
    return Cook_D
```

Question 2

Point out the differences between H-matrix and Cook's distance in the algorithm from a theoretical point of view. Mainly, you should be able to verify...

- Are the outliers identified by the two methods intersect?

Answer

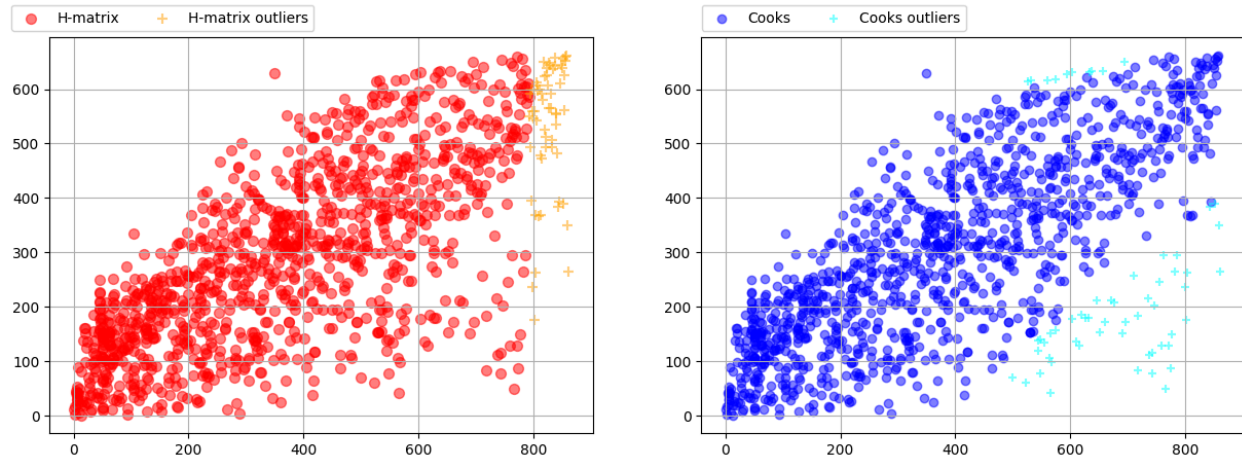


Fig 1: Comparison of observations and detected outliers using **H-matrix** (left) and **Cook's distance** (right) under hold-out (80-20) criteria.

Yes, the outliers identified by the **H-matrix** and **Cook's distance** partially intersect but are not identical. Observations with **both high leverage and large residuals** are detected by both methods.

Since **Cook's distance** incorporates residuals and MSE of y in its formulation, it identifies only **influential observations** that significantly affect the fitted model. In contrast, the **H-matrix** considers only leverage derived from the feature space (X), and may therefore identify points with high leverage.

Consequently, **Cook's distance** can be regarded as a more restrictive subset of **H-matrix outliers**, focusing on points with a substantial impact on the regression model.

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- b. Are there any outliers identified that have been chosen differently by each of the methods?

Answer

Yes, the two methods identify outliers **differently due to their selection criteria**. The **H-matrix** identifies observations with **high leverage**, meaning they are extreme in the feature space (X), even if their residuals are small. These points are therefore **selected only by the H-matrix** and not by Cook's distance.

In contrast, **Cook's distance** identifies only **influential observations**, requiring both **high leverage and large residuals**.

As a result, the outliers detected by Cook's distance form a **more restrictive subset** of those identified by the H-matrix.

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- c. How is the **performance improved** by removing the outliers identified by H-matrix versus the Cook's distance?

Answer

Performance improvement from removing outliers identified by the **H-matrix** and **Cook's distance** arises from different mechanisms. The H-matrix detects **high-leverage points in the input space**, which may not have large residuals but can distort the regression geometry. Removing such points stabilizes $X @ X.T$, reduces the variance of parameter estimates, and often improves **generalization performance**. In contrast, Cook's distance identifies observations with **both high leverage and large residuals**, i.e., points that strongly influence the fitted coefficients. Removing these observations directly reduces bias in parameter estimation and typically leads to a **significant reduction in training MSE**. Therefore, H-matrix-based removal improves model stability, while Cook's distance-based removal improves the fit by eliminating highly influential observations.